

Metasurface RCWA Simulation Project

This repository contains simulation and analysis tools for studying nanoscale metasurface spectral filters using Rigorous Coupled-Wave Analysis (RCWA). The goal is to reproduce and explore periodic nanostructured “code mask” designs that control optical transmission spectra for applications in computational optics and nanophotonic signal encoding.

Project Structure

`rcw_grad/` – RCWA simulation engine (external repository)
`scripts/` – Custom simulation and analysis scripts
`geometry/` – Unit cell and metasurface design definitions
`results/` – Generated transmission spectra and data outputs
`figures/` – Plots and visualizations
`notes/` – Documentation and research notes

Research Goals

- Reproduce metasurface spectral filter results from recent literature.
- Study the relationship between unit cell geometry and transmission spectra.
- Build a library of simulated optical responses for different structures.

Methods

The project utilizes **Rigorous Coupled-Wave Analysis (RCWA)** to solve Maxwell's equations for periodic structures. Key workflows include:

- Parameter sweeps over geometric variables (p , a , b , d).
- Spectral transmission analysis across a wide wavelength range.
- Gradient-based optimization of unit cell parameters.

Project Status

Initial setup and baseline simulation reproduction in progress.